MIL-S-13387 (Ord)
20 April 1954
SUPERSEDING
FXS-991
17 May 1945

# MILITARY SPECIFICATION
# SIGHT, REFLEX, M18

## 1. SCOPE

1.1 This specification covers a reflex sight used as a tracking aid with 40 MM., AA Weapons or direct fire against aircraft.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

| | |
|---|---|
| JAN-F-675 | Films, Reflection-Reducing for Glass Optical Elements. |
| MIL-P-11946 | Packaging of Sight, Reflex M18 |

U. S. Army

| | |
|---|---|
| 51-7 | Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection of |
| 51-70-2 | Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of |

DRAWINGS

Ordnance Corps

Class 91, Division 258, Drawing 1, Sight, Reflex, M18

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

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## 3. REQUIREMENTS

3.1 General.- Unless specified otherwise, the requirements common to all fire control instruments as specified in Specification 51-7 and Specification 51-70-2 shall be mandatory insofar as they are applicable.

3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Vibration.- After being subjected to the vibration test of Specification 51-70-2, the instrument shall show no evidence of dirt when viewed through the objective. The vibration test shall be the first test performed on the instrument.

3.4 Position of Reticle.- With the axis of the mount assembly C7675757 horizontal, as shown on drawing D7693787, the image of the reticle shall have the spaces that form a cross on the reticle alined within 5 degrees in both the horizontal and vertical positions.

3.5 Image Quality.- The image of the reticle shall be clear and uniform without any break or irregularities.

3.6 Reticle Illumination.- With the illuminator in place and energized, the reticle image shall appear clearly defined when viewed against skylight or a background with a brightness of 300 foot candles

3.7 Parallax.- Parallax between the reticle image and a distant target (greater than 100 yards) shall not exceed 0.2 mil.

3.8 Adjustment.- The azimuth and elevation adjustment screws shall be set to the approximate center of their respective movements when the instrument is bore-sighted.

3.9 Abrasion Resistance.- There shall be no scratches larger than size 40; the combined length of all scratches size 40 and smaller shall not exceed 3 inches in the rubbed area of a reflector surface which has been subjected to the abrasion test specified in 4.4.

3.10 Weatherproofing.- The reflex sight shall be sealed properly against the entrance of rain into its interior.



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3.11 Temperature.- Temperature requirements specified in Specification 51-7 for Type II instruments shall apply.

3.12 Workmanship.- Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- Unless specified otherwise, each sight shall be inspected for conformance with this specification and with Specification 51-7 and Specification 51-70-2. Unless specified otherwise, standard ambient tests shall be performed at a temperature of 60°F to 90°F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Place of inspection.- Unless specified otherwise by the contracting officer, inspection and tests shall be conducted at the place of manufacture.

4.3 Vibration Test.- The vibration test specified in Specification 51-70-2 shall apply.

4.4 Abrasion Resistance Test.- The abrasion resistance test specified in Specification JAN-P-675 shall apply.

4.5 Weatherproofing Test.- One reflex sight out of every ten shall be subjected to the artificial rain test specified in Specification 51-7. This test shall be applied with the illuminator positioned for daylight illumination.

4.6 Temperature test.- The temperature test contained in Specification 51-7 for Type II instruments shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-11946.

6. NOTES

6.1 Intended use.- Reflex Sight M18 is used as a tracking aid for direct fire against aircraft.

6.2 Ordering data.- Invitation for bids and contracts should specify the title, number, and date of this specification.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8370

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